

GED Science Practice Test 5

PART A — QUESTIONS

GED-SCI5-001

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student investigates whether carbon dioxide affects photosynthesis in aquatic plants. Three identical containers each hold the same amount of water and the same type and mass of aquatic plants. Container A receives no added carbon dioxide, Container B receives a low amount, and Container C receives a high amount. All containers receive the same light for 30 minutes. The student counts oxygen bubbles released by the plants.

Carbon dioxide level | Oxygen bubbles in 30 min

None added | 14

Low | 32

High | 51

Which conclusion is best supported by the data?

- A) Higher carbon dioxide levels were associated with more oxygen production.
- B) Carbon dioxide stopped photosynthesis in all containers.
- C) Plants produced the most oxygen when no carbon dioxide was added.
- D) Light had no role because all containers received the same light.

GED-SCI5-002

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student measures the mass and volume of an unknown solid and compares it with a table of material densities.

Unknown solid:

Mass = 54 g

Volume = 20 mL

Material | Density (g/mL)

Plastic | 1.2

Glass | 2.7

Aluminum | 2.7

Copper | 8.9

Based on the density, which material could the unknown solid be?

- A) Plastic
- B) Glass or aluminum
- C) Copper
- D) Water

GED-SCI5-003

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

The Moon does not make its own light. It reflects sunlight. As the Moon orbits Earth, people on Earth see different amounts of the sunlit half of the Moon. These changing views repeat in a regular cycle.

What causes the phases of the Moon?

- A) The changing position of the Moon as it orbits Earth
- B) Clouds blocking parts of the Moon every night
- C) The Moon producing different amounts of its own light
- D) Earth's seasons changing from summer to winter

GED-SCI5-004

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

When a small blood vessel is cut, platelets gather at the damaged area. Proteins in the blood help form a clot, which slows or stops bleeding. This response helps protect the body from losing too much blood and from pathogens entering through the wound.

Which statement best describes the function of platelets in this passage?

- A) They help form clots at damaged blood vessels.
- B) They digest proteins in the stomach.
- C) They carry genetic information from parents.
- D) They exchange gases in the lungs.

GED-SCI5-005

Section: Science

Type: numeric_entry

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A wave travels 150 meters in 5 seconds.

What is the speed of the wave in meters per second?

Type your answer as a whole number.

GED-SCI5-006

Section: Science

Type: multi_select

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

Select TWO answers.

A lake is monitored before and after a nearby parking lot is built.

Measurement | Before construction | After construction

Runoff entering lake after storms (m³) | 120 | 210

Nitrate level (mg/L) | 1.8 | 3.4

Average water clarity (m) | 4.2 | 2.5

Which TWO statements are best supported by the data?

- A) Runoff entering the lake increased after construction.
- B) Nitrate level increased after construction.
- C) Water clarity increased after construction.
- D) The parking lot caused the lake to completely dry up.
- E) Nitrate level stayed exactly the same after construction.

GED-SCI5-007

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A biologist counts two species in a grassland over four years.

Year | Field mice counted | Owls counted

1 | 240 | 18

2 | 310 | 25

3 | 180 | 14

4 | 260 | 21

Which conclusion is best supported by the data?

- A) Owl counts tended to be higher in years when field mouse counts were higher.
- B) Owl counts were highest when field mice were lowest.
- C) Field mice disappeared by Year 4.
- D) Owls do not depend on food sources in the grassland.

GED-SCI5-008

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student tests how distance affects the force between a magnet and a paper clip. The student places the magnet at different distances from the paper clip and measures the force needed to pull the paper clip away. The same magnet and paper clip are used in every trial.

What is the independent variable?

- A) Distance between the magnet and paper clip
- B) Force measured
- C) Type of paper clip
- D) Type of magnet

GED-SCI5-009

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A cold front forms when a cold air mass moves into an area occupied by warmer air. The colder, denser air pushes under the warmer air, forcing the warmer air upward. As the warm air rises, water vapor may condense and form clouds or storms.

Which result is most likely when a cold front passes through an area?

- A) Rising warm air may produce clouds and storms.
- B) Ocean tides stop for several days.
- C) Earth's rotation reverses direction.
- D) Rock layers become fossils immediately.

GED-SCI5-010

Section: Science

Type: dropdown

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Some molecules move across a cell membrane from an area of higher concentration to an area of lower concentration without using cellular energy. This process is [D1]. Movement that requires cellular energy is [D2].

Dropdowns:

D1) passive transport | mitosis | digestion

D2) active transport | evaporation | sedimentation

GED-SCI5-011

Section: Science

Type: drag_drop

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Match each target with the correct tile.

Tiles:

- A) Length of the pendulum string
- B) Time for 10 complete swings
- C) Same mass attached to the string
- D) Color of the stopwatch

Targets:

- T1) Independent variable
- T2) Dependent variable
- T3) Controlled variable

GED-SCI5-012

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The endocrine system uses hormones to send chemical messages through the bloodstream. One hormone, adrenaline, can increase heart rate and breathing rate during a stressful situation. These changes help deliver more oxygen and glucose to muscles.

Which statement best describes the role of hormones in this passage?

- A) Hormones act as chemical messages that affect body functions.
- B) Hormones break bones into smaller pieces.
- C) Hormones are gases exchanged in the lungs.
- D) Hormones prevent blood from moving through vessels.

GED-SCI5-013

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student tears a sheet of paper into smaller pieces. The paper has a different shape, but it is still paper. Later, the student burns another sheet of paper, producing ash, smoke, and gases.

Which statement correctly compares the two changes?

- A) Tearing is a physical change, while burning is a chemical change.
- B) Tearing is a chemical change, while burning is a physical change.
- C) Both tearing and burning are only changes in shape.
- D) Neither process changes the paper in any way.

GED-SCI5-014

Section: Science

Type: numeric_entry

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A river deposits 48 centimeters of sediment in 12 years.

What is the average sediment deposition rate in centimeters per year?

Type your answer as a whole number.

GED-SCI5-015

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher tests whether a new plant spray reduces damage from leaf-eating insects. Twenty similar plants are divided into two groups. One group receives the spray, and the other receives plain water. Both groups are placed in the same greenhouse and exposed to the same number of insects. After one week, the researcher measures the area of leaves eaten.

Which evidence would best support the claim that the spray works?

- A) Sprayed plants have less leaf area eaten on average.

- B) Water-treated plants have less leaf area eaten on average.
- C) Both groups have exactly the same leaf damage.
- D) The insects are counted before being placed in the greenhouse.

GED-SCI5-016

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student records the position of a cart moving in a straight line.

Time (s): 0, 2, 4, 6

Position (m): 0, 5, 10, 15

Which statement best describes the cart's motion?

- A) The cart moves at a constant speed of 2.5 m/s.
- B) The cart moves at a constant speed of 5 m/s.
- C) The cart is not moving.
- D) The cart moves backward after 2 seconds.

GED-SCI5-017

Section: Science

Type: dropdown

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

When water evaporates from the ocean, salt is left behind. Later, water vapor cools and forms clouds through [D1]. Water that falls from clouds to Earth is called [D2].

Dropdowns:

D1) condensation | combustion | subduction

D2) precipitation | magma | radiation

GED-SCI5-018

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A hospital tracks the number of patients with a certain infection before and after a new cleaning protocol is used.

Month | Patients with infection

Before Month 1 | 24

Before Month 2 | 22

After Month 1 | 13

After Month 2 | 11

Which claim is best supported by the data?

- A) The number of patients with the infection decreased after the protocol began.
- B) The number of patients with the infection doubled after the protocol began.

- C) The cleaning protocol caused every infection to disappear immediately.
- D) Infection numbers stayed exactly the same across all months.

GED-SCI5-019

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student wants to compare how different surfaces affect sliding friction. The student pulls the same wooden block across sandpaper, plastic, and tile using a spring scale. The block's mass and pulling speed are kept the same for every trial.

What is the dependent variable?

- A) Type of surface
- B) Force measured by the spring scale
- C) Mass of the wooden block
- D) Pulling speed

GED-SCI5-020

Section: Science

Type: multi_select

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

Select TWO answers.

A class measures the average mass gain of young animals fed different diets for four weeks.

Diet | Average mass gain (g)

Standard diet | 38

High-protein diet | 52

Low-protein diet | 21

High-sugar diet | 35

Which TWO statements are best supported by the data?

- A) The high-protein diet group had the greatest average mass gain.
- B) The low-protein diet group had the smallest average mass gain.
- C) The high-sugar diet group gained more mass than the high-protein group.
- D) All diets produced the same average mass gain.
- E) The standard diet group gained less mass than the low-protein group.

GED-SCI5-021

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

Over many years, waves strike a rocky shoreline. Cracks in the rock widen, and pieces break off. The broken pieces are moved by water and deposited farther down the coast.

Which sequence best describes these processes?

- A) Weathering, erosion, and deposition
- B) Condensation, freezing, and orbiting
- C) Photosynthesis, respiration, and digestion
- D) Magnetism, conduction, and reflection

GED-SCI5-022

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student tests whether salt lowers the temperature at which ice melts. The student places equal masses of ice into four cups. Each cup receives a different mass of salt. The cups are kept in the same room, and the student records the lowest temperature reached by the ice-water mixture.

Which factor should be kept constant?

- A) Mass of ice in each cup
- B) Mass of salt added
- C) Lowest temperature reached
- D) Melting result

GED-SCI5-023

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A species of wildflower has two petal colors: blue and white. Bees visit blue flowers more often because blue flowers are easier for them to detect. Blue-flowered plants produce more seeds on average. After many generations, blue flowers become more common.

Which explanation best fits this change?

- A) Blue-flowered plants had a reproductive advantage because pollinators visited them more often.
- B) White flowers changed color because they wanted more bee visits.
- C) Bees caused all wildflowers to stop reproducing.
- D) Petal color cannot be affected by reproductive success.

GED-SCI5-024

Section: Science

Type: numeric_entry

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A sample contains 240 cells. Of these cells, 72 are in the process of dividing.

What percent of the cells are dividing?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI5-025

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A pot of soup is heated on a stove. Warm liquid near the bottom becomes less dense and rises.

Cooler liquid near the top sinks. This circulation helps transfer thermal energy throughout the soup.

Which method of heat transfer is described?

- A) Convection
- B) Reflection
- C) Static discharge
- D) Nuclear fusion

GED-SCI5-026

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student wants to test whether light color affects the rate of photosynthesis in spinach leaves. Leaf disks are placed in identical cups of solution under red, blue, and green lights. The student records how many leaf disks float after 15 minutes. The distance between each lamp and cup is kept the same.

Why should the lamp distance be kept the same?

- A) To keep light intensity similar while testing light color
- B) To make all light colors turn white
- C) To prevent the leaves from containing chloroplasts
- D) To make the solution change into a solid

GED-SCI5-027

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A city records the number of days above 35°C during four summers.

Summer | Days above 35°C

- 1 | 9
- 2 | 14
- 3 | 18
- 4 | 23

What is the increase in the number of days above 35°C from Summer 1 to Summer 4?

- A) 9 days
- B) 14 days
- C) 23 days
- D) 32 days

GED-SCI5-028

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares the wavelengths of four sound waves traveling through the same medium.

Wave | Wavelength (m)

A | 0.5

B | 1.0

C | 1.5

D | 2.0

If the waves travel at the same speed, which wave has the highest frequency?

A) Wave A

B) Wave B

C) Wave C

D) Wave D

GED-SCI5-029

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

In humans, DNA is stored mainly in the nucleus of most body cells. DNA contains instructions for making proteins, and proteins affect many traits and body functions.

Which statement best describes DNA?

A) DNA contains inherited instructions used by cells.

B) DNA is a mineral that strengthens bones.

C) DNA is a gas used directly for breathing.

D) DNA is produced only by the stomach.

GED-SCI5-030

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student measures the mass of reactants before a reaction and the mass of products after the reaction in a sealed container.

Trial | Mass before reaction (g) | Mass after reaction (g)

1 | 80 | 80

2 | 125 | 125

3 | 210 | 210

Which statement is best supported by the data?

A) Mass was conserved in each sealed reaction.

- B) Mass disappeared in each reaction.
- C) The products always had less mass than the reactants.
- D) The container changed mass into energy only.

GED-SCI5-031

Section: Science

Type: multi_select

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student investigates how water temperature affects the time needed for a bath tablet to dissolve. The student uses one identical tablet in each cup and measures dissolving time.

Which TWO steps would improve the investigation?

- A) Use the same volume of water in each cup.
- B) Repeat each temperature trial several times.
- C) Use a different tablet mass for each temperature.
- D) Record only the fastest dissolving time.
- E) Change the cup size and water temperature together.

GED-SCI5-032

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A researcher counts the number of stomata, tiny openings for gas exchange, on leaves from plants grown in dry and moist conditions.

Condition | Average stomata per mm²

Dry | 88

Moist | 142

Which statement is best supported by the data?

- A) Plants grown in moist conditions had more stomata per mm² on average.
- B) Plants grown in dry conditions had more stomata per mm² on average.
- C) Stomata were absent in both conditions.
- D) Moisture conditions had no relationship to stomata count.

GED-SCI5-033

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A glass prism separates white light into different colors. This happens because different wavelengths of visible light bend by different amounts as they pass from air into glass and back into air.

Which wave property is mainly involved in separating the colors?

- A) Wavelength
- B) Mass
- C) Density of sound
- D) Electrical resistance

GED-SCI5-034

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A population of bacteria is exposed to a cleaning chemical. Most bacteria die, but a few survive because they already have traits that help them tolerate the chemical. The surviving bacteria reproduce, and the tolerant trait becomes more common in later generations.

Which conclusion is best supported?

- A) The cleaning chemical selected for bacteria with tolerance traits.
- B) The bacteria stopped passing traits to offspring.
- C) The chemical made all bacteria identical immediately.
- D) Tolerant bacteria survived because they did not contain DNA.

PART B — ANSWER KEY + EXPLANATIONS

GED-SCI5-001 — Correct: A — Correct Answer: Higher carbon dioxide levels were associated with more oxygen production. — Explanation: Oxygen bubble counts increased as carbon dioxide level increased, supporting the conclusion that more carbon dioxide was associated with more photosynthesis.

GED-SCI5-002 — Correct: B — Correct Answer: Glass or aluminum — Explanation: Density = $54 \text{ g} \div 20 \text{ mL} = 2.7 \text{ g/mL}$, which matches the listed density for glass and aluminum.

GED-SCI5-003 — Correct: A — Correct Answer: The changing position of the Moon as it orbits Earth — Explanation: Moon phases are caused by the Moon's changing position relative to Earth and the Sun.

GED-SCI5-004 — Correct: A — Correct Answer: They help form clots at damaged blood vessels. — Explanation: Platelets gather at damaged blood vessels and help form clots.

GED-SCI5-005 — Correct: 30 — Tolerance: exact — Correct Answer: 30 — Explanation: Speed = distance $\div$ time = $150 \text{ m} \div 5 \text{ s} = 30 \text{ m/s}$.

GED-SCI5-006 — Correct: A,B — Correct Answer: Runoff entering the lake increased after construction. | Nitrate level increased after construction. — Explanation: Runoff increased from 120 m^3 to 210 m^3 , and nitrate level increased from 1.8 mg/L to 3.4 mg/L .

GED-SCI5-007 — Correct: A — Correct Answer: Owl counts tended to be higher in years when field mouse counts were higher. — Explanation: Years with more field mice generally also had more owls, suggesting a predator-prey relationship.

GED-SCI5-008 — Correct: A — Correct Answer: Distance between the magnet and paper clip — Explanation: The independent variable is the factor intentionally changed by the student.

GED-SCI5-009 — Correct: A — Correct Answer: Rising warm air may produce clouds and storms. — Explanation: A cold front forces warm air upward, which can lead to condensation, clouds, and storms.

GED-SCI5-010 — Correct: D1=passive transport; D2=active transport — Correct Answer: D1=passive transport; D2=active transport — Explanation: Passive transport does not require cellular energy, while active transport does require energy.

GED-SCI5-011 — Correct: T1=A; T2=B; T3=C — Correct Answer: T1=Length of the pendulum string; T2=Time for 10 complete swings; T3=Same mass attached to the string — Explanation: The string length is changed, the swing time is measured, and the attached mass is kept constant.

GED-SCI5-012 — Correct: A — Correct Answer: Hormones act as chemical messages that affect body functions. — Explanation: Hormones travel through the bloodstream and help regulate body processes.

GED-SCI5-013 — Correct: A — Correct Answer: Tearing is a physical change, while burning is a chemical change. — Explanation: Tearing changes size and shape, while burning forms new substances such as ash and gases.

GED-SCI5-014 — Correct: 4 — Tolerance: exact — Correct Answer: 4 — Explanation: Deposition rate = $48 \text{ cm} \div 12 \text{ years} = 4 \text{ cm/year}$.

GED-SCI5-015 — Correct: A — Correct Answer: Sprayed plants have less leaf area eaten on average. — Explanation: Less leaf damage in the sprayed group would support the claim that the spray reduces insect damage.

GED-SCI5-016 — Correct: A — Correct Answer: The cart moves at a constant speed of 2.5 m/s. — Explanation: The cart moves 5 meters every 2 seconds, so speed = $5 \div 2 = 2.5 \text{ m/s}$.

GED-SCI5-017 — Correct: D1=condensation; D2=precipitation — Correct Answer: D1=condensation; D2=precipitation — Explanation: Condensation forms clouds, and precipitation is water falling from clouds to Earth.

GED-SCI5-018 — Correct: A — Correct Answer: The number of patients with the infection decreased after the protocol began. — Explanation: Infection counts were lower in the months after the new cleaning protocol began.

GED-SCI5-019 — Correct: B — Correct Answer: Force measured by the spring scale — Explanation: The dependent variable is the measured outcome in response to changing the surface type.

GED-SCI5-020 — Correct: A,B — Correct Answer: The high-protein diet group had the greatest average mass gain. | The low-protein diet group had the smallest average mass gain. — Explanation: The high-protein group had the highest average gain, 52 g, and the low-protein group had the lowest, 21 g.

GED-SCI5-021 — Correct: A — Correct Answer: Weathering, erosion, and deposition — Explanation: Rock breaks apart by weathering, pieces are moved by erosion, and they are deposited elsewhere.

GED-SCI5-022 — Correct: A — Correct Answer: Mass of ice in each cup — Explanation: Keeping ice mass constant helps isolate the effect of salt amount.

GED-SCI5-023 — Correct: A — Correct Answer: Blue-flowered plants had a reproductive advantage because pollinators visited them more often. — Explanation: Plants visited more often by pollinators produced more seeds, making blue flowers more common over generations.

GED-SCI5-024 — Correct: 30 — Tolerance: exact — Correct Answer: 30 — Explanation: $72 \div 240 \times 100 = 30\%$.

GED-SCI5-025 — Correct: A — Correct Answer: Convection — Explanation: Convection transfers thermal energy through the movement of warmer and cooler fluid.

GED-SCI5-026 — Correct: A — Correct Answer: To keep light intensity similar while testing light color — Explanation: Keeping lamp distance constant helps ensure that light color is the main variable being tested.

GED-SCI5-027 — Correct: B — Correct Answer: 14 days — Explanation: The increase is $23 - 9 = 14$ days.

GED-SCI5-028 — Correct: A — Correct Answer: Wave A — Explanation: For waves traveling at the same speed, shorter wavelength means higher frequency. Wave A has the shortest wavelength.

GED-SCI5-029 — Correct: A — Correct Answer: DNA contains inherited instructions used by cells. — Explanation: DNA stores genetic instructions used by cells to make proteins and influence traits.

GED-SCI5-030 — Correct: A — Correct Answer: Mass was conserved in each sealed reaction. — Explanation: In each trial, the mass before and after the reaction is the same, supporting conservation of mass.

GED-SCI5-031 — Correct: A,B — Correct Answer: Use the same volume of water in each cup. | Repeat each temperature trial several times. — Explanation: Keeping water volume constant controls the setup, and repeating trials improves reliability.

GED-SCI5-032 — Correct: A — Correct Answer: Plants grown in moist conditions had more stomata per mm^2 on average. — Explanation: The moist-condition leaves had 142 stomata per mm^2 compared with 88 in dry conditions.

GED-SCI5-033 — Correct: A — Correct Answer: Wavelength — Explanation: Different wavelengths of visible light bend by different amounts in a prism, separating white light into colors.

GED-SCI5-034 — Correct: A — Correct Answer: The cleaning chemical selected for bacteria with tolerance traits. — Explanation: Bacteria with tolerance traits survived and reproduced, making the trait more common over generations.